2.6 Warm-up: Mental Math Solutions

Students write only the answer. The notes give the step a student should be doing in their head, and the mistakes worth looking for.

1. Fill in the blanks.

   (a) 3.2 km = **3200 m**
   (b) 60 cm = **0.6 m**
   (c) 1.5 m = **150 cm**
   (d) 9 mm = **0.9 cm**
   (e) 0.4 m = **400 mm**
   (f) 750 m = **0.75 km**

   Going to a smaller unit makes the number bigger. For (f), going to a bigger unit, divide by 1000: 750 m is three quarters of a kilometer.

2. Squares and square roots.

   (a) $4^2 = \mathbf{16}$
   (b) $9^2 = \mathbf{81}$
   (c) $11^2 = \mathbf{121}$
   (d) $30^2 = \mathbf{900}$
   (e) $0.2^2 = \mathbf{0.04}$
   (f) $1.5^2 = \mathbf{2.25}$
   (g) $\sqrt{49} = \mathbf{7}$
   (h) $\sqrt{400} = \mathbf{20}$

   The traps: $0.2^2 = 0.2 \times 0.2 = 0.04$*, not 0.4, and* $30^2 = 900$*, not 90. For (f),* $1.5 \times 1.5 = 1.5 + 0.75$*. For (h),* $20 \times 20 = 400$*.*

3. Speed units.

   (a) 5 m/s = **18 km/h**
   (b) 15 m/s = **54 km/h**
   (c) 72 km/h = **20 m/s**
   (d) 90 km/h = **25 m/s**

   From m/s to km/h, multiply by 3.6; back again, divide. 72 is 2×36*, and 90 is* 2.5×36*. The number in km/h is always the bigger one.*

4. Distance = speed × time.

   (a) **30 m** $1.5 \times 20 = 30$.
   (b) **360 m** One minute is 60 s, and $6 \times 60 = 360$.
   (c) **30 km** 20 minutes is $\frac{1}{3}$ hour, and $90 \times \frac{1}{3} = 30$.
   (d) **20 s** time = distance ÷ speed = $100 \div 5$.

   In (b) and (c) the time has to be in the same unit as the speed first. A student who writes 6 m or 1800 km has skipped that step.

5. One lap of the track.

   Average speed = **5 m/s** $(400 \div 80)$ Average velocity = **0** (displacement is zero)

   The runner finishes where the lap began, so the displacement is 0, and so is the average velocity, however fast the runner went.

6. Fractions as decimals.

   (a) $\frac{3}{5} = \mathbf{0.6}$ (c) $\frac{3}{20} = \mathbf{0.15}$ (e) $\frac{3}{25} = \mathbf{0.12}$ (g) $\frac{1}{9} = \mathbf{0.\overline{1}}$

   (b) $\frac{9}{10} = \mathbf{0.9}$ (d) $\frac{1}{50} = \mathbf{0.02}$ (f) $\frac{5}{4} = \mathbf{1.25}$ (h) $\frac{4}{3} = \mathbf{1.\overline{3}}$

   Think in money: $\frac{3}{20}$ *is three nickels,* $\frac{1}{50}$ *is 2 cents,* $\frac{3}{25}$ *is 12 cents. Accept* $0.111\ldots$ *and* $1.333\ldots$*, or 0.11 and 1.33 written with* $\approx$*.*

7. Fractions of an hour and a week.

   (a) 0.5 hour = **30 min**

   (b) 0.4 hour = **24 min**

   (c) 2.5 min = **150 s**

   (d) 90 s = **1.5 min**

   (e) 6 min = **0.1 h**

   (f) 40 min = $\mathbf{\frac{2}{3}}$ **h**

   (g) 1 week = **168 hours**

   (h) 2.25 hours = **2 h 15 min**

   Each tenth of an hour is 6 minutes. For (g), $7 \times 24 = 140 + 28$*. In (h), 0.25 h is 15 min, not 25 min.*

8. Acceleration = change in velocity ÷ time.

   (a) **4 m/s²** $20 \div 5$.

   (b) **2 m/s²** The change is $8 - 2 = 6$ m/s, and $6 \div 3 = 2$. *The slip is* $8 \div 3$*, using the final speed instead of the change.*

   (c) **4 s** $20 \div 5$.

   (d) Average velocity = **10 m/s** $(\frac{0+20}{2})$; distance = **50 m** (10×5).

9. Falling with $g \approx 10$ m/s².

   (a) **30 m/s** 10×3.

   (b) **20 m** $\frac{1}{2} \times 10 \times 2^2 = 5 \times 4$. *Not 10 m: the stone is only going 20 m/s at the end, so its average is 10 m/s.*

   (c) **2 s** It loses 10 m/s every second, and $20 \div 10 = 2$.

   (d) Velocity = **0**; acceleration = **10 m/s² downward**.

   The common wrong answer is $a = 0$ *at the top. Gravity does not switch off; the ball is still slowing down, then speeding up downward.*

10. **About 26 miles.** $42 \times \frac{5}{8} = \frac{210}{8} = 26\frac{1}{4}$.

    Accept 25 to 26.25 mi. The marathon is 42.195 km, which is 26.2 mi.